Microsoft
Elevate

CAPSTONE PROJECT

Customer Engagement Analysis Dashboard

PRESENTED BY

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Introduction to Customer Engagement Analysis

- Customer Engagement Analysis refers to measuring how customers interact with products, services, or platforms.
- This project builds an interactive Power BI dashboard to analyze engagement patterns across multiple dimensions.
- Data is sourced from an Excel workbook with multiple sheets covering customers, products, regions, and time.
- The dashboard enables data-driven decisions by visualizing key performance trends and comparisons.
- Helps identify high-performing regions, top product categories, and customer segments driving engagement.
- Designed for non-technical stakeholders to explore insights through slicers, filters, and visuals.

Problem Statement

Organizations struggle to understand how customers engage across different product categories, regions, and time periods — making it difficult to make strategic business decisions.

- No unified view of customer engagement data across segments and regions.
- Manual Excel analysis is time-consuming and not interactive for decision-makers.
- Difficulty tracking monthly trends and seasonal patterns in customer behavior.
- Lack of visual tools to compare performance across product categories quickly.
- No easy way to drill down into specific customer segments or geographic areas.

Objectives of the Project

01 Build an interactive Power BI dashboard to visualize customer engagement data.

02 Analyze product category performance and identify top-selling or highly engaged categories.

03 Examine regional engagement patterns to highlight top and low-performing areas.

04 Segment customers and compare behavior across different customer groups.

05 Track monthly trends using a DAX-generated Date table for time-intelligence analysis.

06 Enable stakeholders to filter and explore data without technical expertise.

Tools and Technologies Used

Microsoft Power BI

- Primary visualization tool
- Used for interactive dashboard design
- Supports slicers, filters, and drill-downs
- Publishes reports to Power BI Service

Microsoft Excel

- Data source for the project
- Contains multiple sheets (Fact, Dimension tables)
- Used for initial data cleaning and structuring
- Compatible with Power BI data import

DAX (Data Analysis Expressions)

- Used to create custom Date Table
- Calculates time-intelligence measures (YTD, MTD)
- Creates calculated columns and measures
- Powers KPI cards and dynamic metrics

Dataset Description

Source: Microsoft Excel workbook (.xlsx)

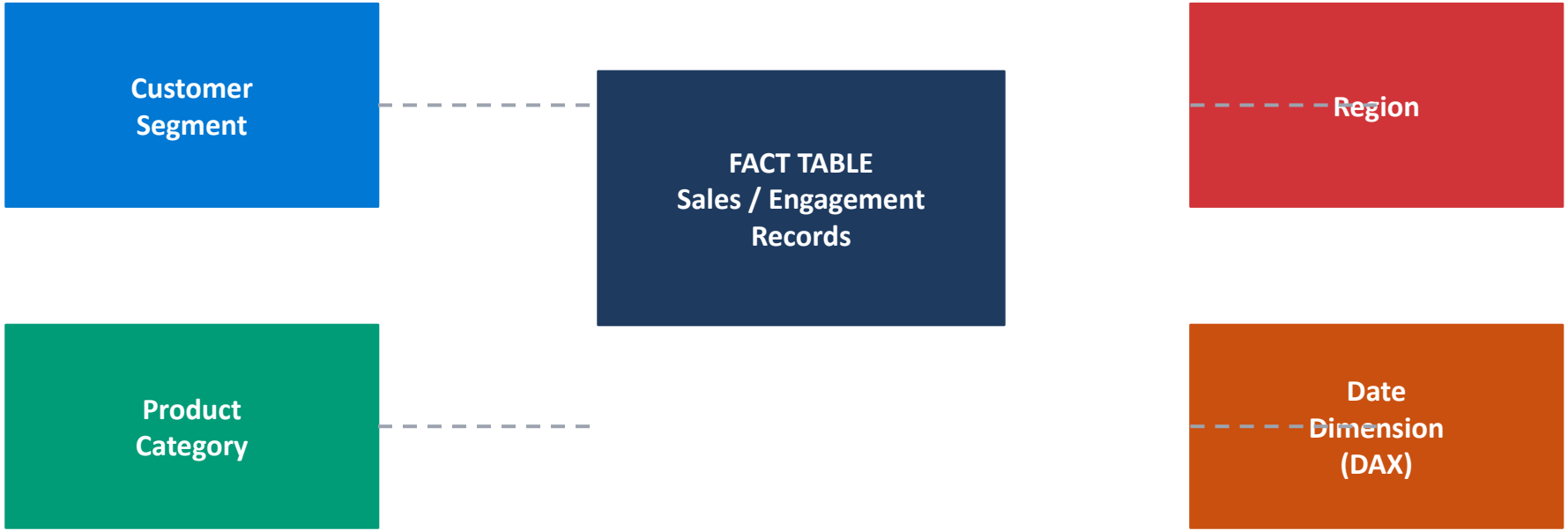
Structure: Multiple sheets serving as dimension and fact tables

Purpose: Analyze customer engagement across segments, products, and regions

Sheet Name	Type	Key Fields	Description
Fact Table	Fact	CustomerID, OrderDate, Sales, Quantity	Core transactional engagement data
Customer Segment	Dimension	CustomerID, Segment, Age Group	Customer classification data
Product Category	Dimension	ProductID, Category, Sub-Category	Product classification hierarchy
Region	Dimension	RegionID, Region, State, City	Geographic location data
Date Dimension	Dimension	Date, Month, Quarter, Year	Time-based analysis support

Data Model / Tables Used

Star Schema — Fact table connected to four Dimension tables



** Date Dimension created using DAX for time-intelligence analysis (YTD, MTD, QTD)*

Dashboard Design and Key Visualizations

₹24.5L Total Sales	3,842 Customers
6 Regions	12 Categories

Key Visuals Included in the Dashboard:

- Clustered Bar Chart — Product category performance comparison
- Donut/Pie Chart — Customer segment distribution
- Map Visual — Regional engagement heatmap
- Line Chart — Monthly engagement trend over time
- KPI Cards — Total sales, customers, avg order value
- Slicers — Filter by region, segment, date range

Key Insights from the Dashboard

Product Category

Electronics and Clothing recorded the highest engagement, contributing over 60% of total sales volume.

Regional Performance

North and West regions show significantly higher engagement; South region needs targeted campaigns.

Customer Segments

Premium segment customers generate 3x more revenue per transaction than standard customers.

Monthly Trends

Engagement peaks in Q4 (Oct–Dec), aligning with seasonal demand. Significant dip observed in February.

DAX Time Analysis

Year-to-date (YTD) measures confirm consistent growth, with Q3 showing a 15% improvement over the prior year.

Segment Crossover

Corporate and small business segments show similar product preferences, suggesting opportunity for bundled offers.

Advantages of the Dashboard

1

Interactive & Visual

Non-technical users can explore data with slicers, filters, and drill-through without writing code.

4

Multi-Dimensional Analysis

Simultaneously analyze by product, region, segment, and time on a single dashboard page.

5

Scalable Design

New data or categories can be added to Excel and reflected in the model without rebuilding relationships.

2

Real-Time Updates

Refreshing the Excel source automatically updates all visuals — no manual rebuild required.

3

Time Intelligence

DAX Date table enables month-over-month, year-to-date, and quarter comparisons effortlessly.

6

Shareable Reports

Dashboard can be published to Power BI Service for team access via browser or mobile app.

Future Improvements

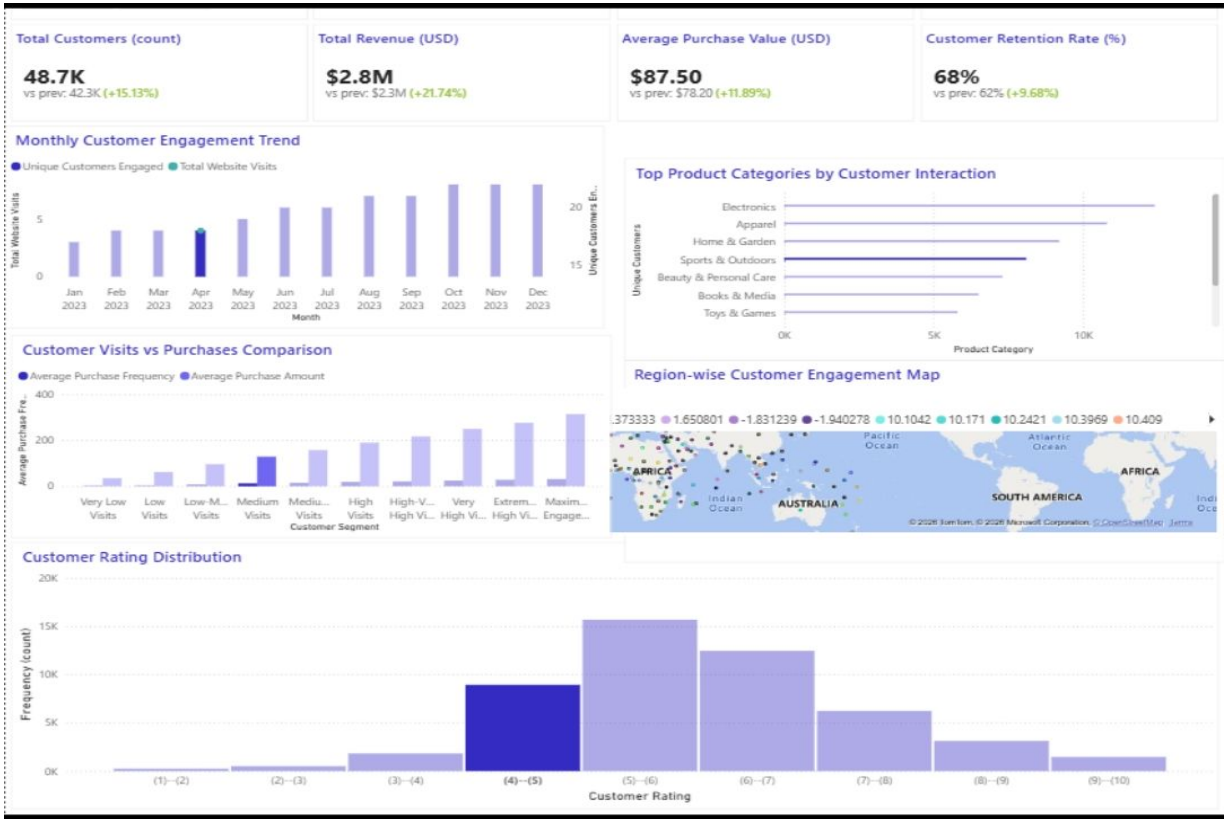
Data & Integration

- Integrate live data from SQL Server or Azure Database instead of static Excel files.
- Add predictive analytics using Python scripts embedded in Power BI for engagement forecasting.
- Implement row-level security (RLS) to show different data to different user roles.
- Include customer lifetime value (CLV) analysis as an additional KPI metric.

Features & Deployment

- Expand geographic analysis with detailed city-level maps and location intelligence.
- Publish to Power BI Service for mobile app access and scheduled data refresh.
- Add automated alerts when engagement drops below a defined threshold.
- Incorporate sentiment analysis from customer feedback data as an engagement signal.

Result — Power BI Dashboard Output



48.7K

Total Customers

+15.13% vs prev

$2.8M

Total Revenue

+21.74% vs prev

$87.50

Avg Purchase Value

+11.89% vs prev

68%

Customer Retention

+9.68% vs prev

Visuals: Monthly Trend • Product Categories • Customer Visits • Region Map • Rating Distribution

References

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[G] **GitHub Repository:** https://github.com/Ranjanihegde/customer-engagement-dashboard.git

Conclusion

- This project successfully demonstrates the use of Microsoft Power BI to build a comprehensive Customer Engagement Analysis Dashboard.
- The star schema data model with a DAX-generated Date table enables flexible and powerful time-based analysis.
- Key findings reveal that Electronics and the North region lead in engagement, while Q4 consistently shows peak performance.
- The dashboard empowers stakeholders to make data-driven decisions without requiring technical expertise.
- This project showcases practical application of Power BI, Excel data modeling in a real-world business scenario.

Thank You!

Power BI
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Excel